Cambridge (CIE) IGCSE Physics
Paper 2: Multiple Choice (Set F)
Extended

Saturday 26 April 2025

Morning (Time: 0 hours 45 minutes)

Total marks

/ 40

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8N (acceleration of free fall = 9.8m/ s^2).

Scan here to mark your mock exam

or visit the mock exams landing page for this course

1 Identify the physical quantity that is a scalar.

A. weight

B. distance

C. velocity

D. acceleration

(1 mark)

2 During a Go Karting race, a car does 8 laps of a 300 m course. It takes 4.0 minutes to complete the race.

What was the average speed of the Go Kart?

A. 75 m/s

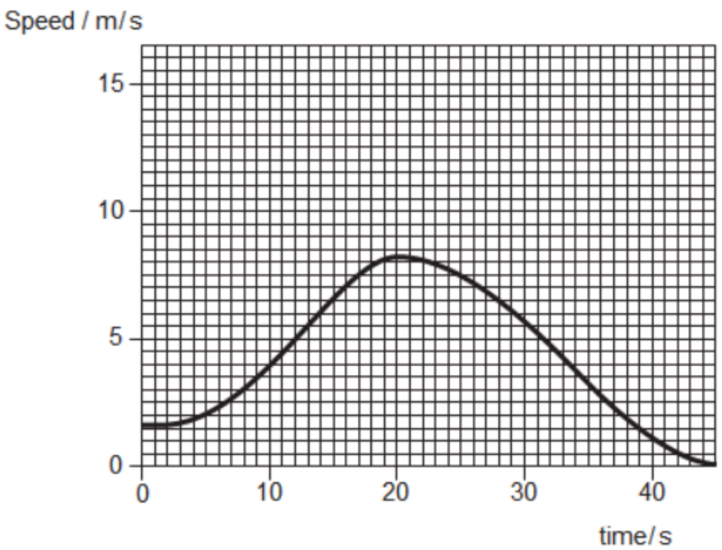
B. 10 m/s

C. 1.25 m/s

D. 600 m/s

(1 mark)

3 The graph shows the speed-time graph of a cyclist moving in a straight line.



What is the acceleration of the cyclist at a time of 20 seconds?

A. 0.5 m/s^2

B. -0.5 m/s^2

C. 0 m/s^2

D. 11.5 m/s^2

(1 mark)

4 A skydiver is falling at terminal velocity. He has not yet opened his parachute.

He opens his parachute.

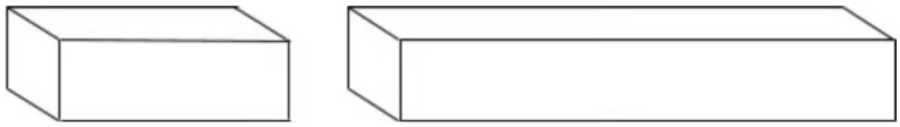
What is the direction of both the velocity of the skydiver, and his acceleration?

	Direction of the skydiver's velocity	**Direction of the skydiver's acceleration**
A	upwards	upwards
B	upwards	downwards
C	downwards	upwards
D	downwards	downwards

(1 mark)

5 Two beams have the same rectangular cross section, but one is longer than the other.

Both beams are made from the same material.



Which quantity is the same in both beams?

A. The volume

B. The weight

C. The mass

D. The density

(1 mark)

6 A car slows down as it approaches a traffic light. The car was traveling at 13.5 m/s and it takes 22 s to come to a stop.

The mass of the car is 950 kg.

Calculate the force required to stop the car.

A. -580 N

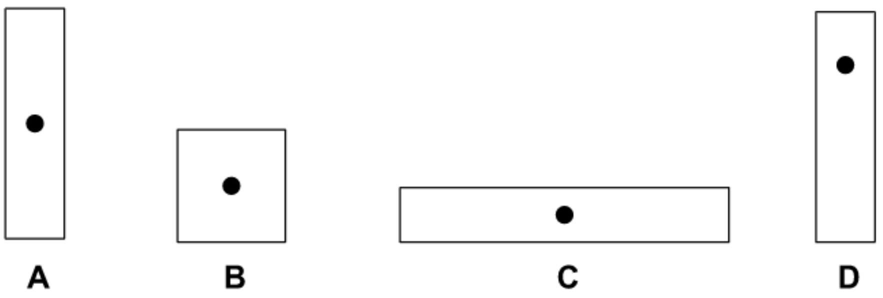
B. 580 N

C. 618 N

D. 1548 N

(1 mark)

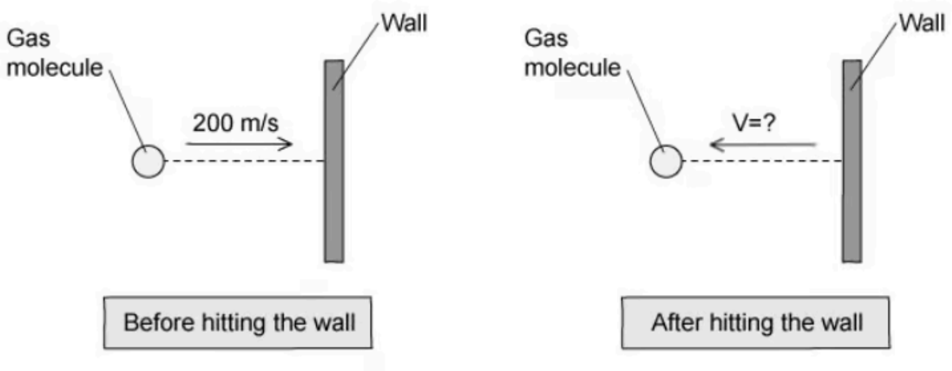
7 Four cuboids are shown in the diagram below. The position of their centre of gravity is also shown.



Which of the cuboids is the most stable?

(1 mark)

8 A gas molecule strikes the wall of a container with a speed of 200m/s. It rebounds with the same kinetic energy as it had before striking the wall.



What is its final velocity?

A. 100 m/s

B. - 100 m/s

C. 200 m/s

D. - 200 m/s

(1 mark)

9 During a paintball fight, a paint pellet of mass 150g hits a stationary target with a speed of 220m/s. It takes 0.025s from the moment the pellet comes into contact with the wall until it flattens onto the wall.

What is the force exerted as a result of the paintball "splat"?

A. 1.3×10^3 N

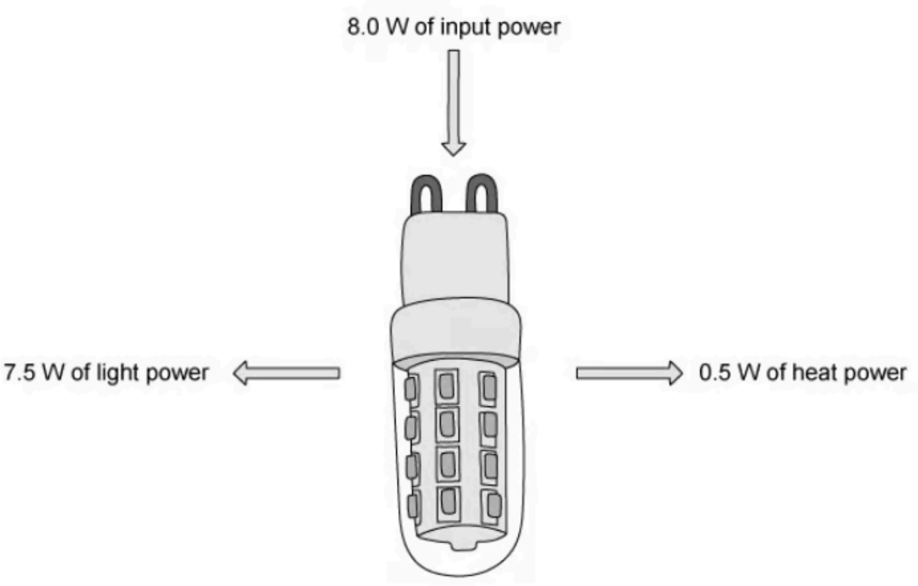
B. 33 N

C. 1.3×10^6 N

D. 33×10^3 N

(1 mark)

10 The diagram shows an LED light bulb.



Which expression gives the efficiency of the bulb?

A. $\frac{8.0}{0.5} \times 100\%$

B. $\frac{7.5}{0.5} \times 100\%$

C. $\frac{0.5}{8.0} \times 100\%$

D. $\frac{7.5}{8.0} \times 100\%$

(1 mark)

11 A trolley is pushed with a force that causes it to move a certain distance. It moves in the direction it is pushed.

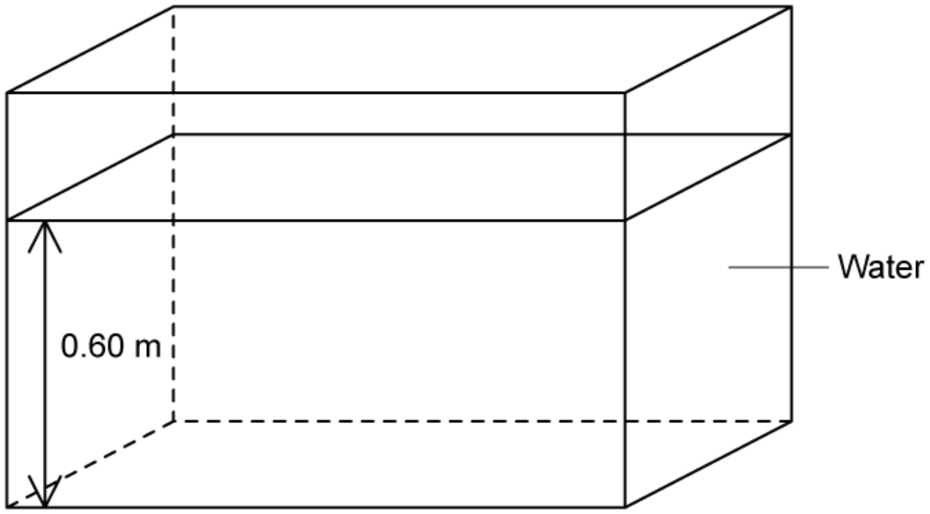
Which row in the table represents the combination of force and distance that means the smallest amount of work is done on the trolley?

	Force / N	**Distance moved / m**
A	20	20
B	40	10
C	50	6.0
D	70	4.0

(1 mark)

12 For the tank of water in the diagram below, which value gives the pressure on the base of the tank due to the water?

The density of water = 1000 kg/m^3



A. 5000 Pa

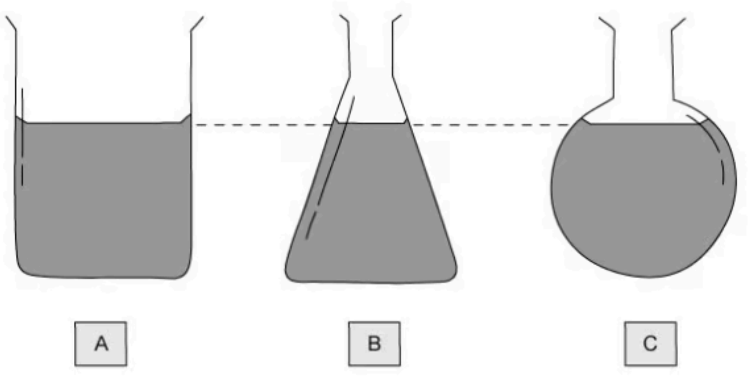
B. 5500 Pa

C. 5880 Pa

D. 6570 Pa

(1 mark)

13 Three beakers of water are placed on a table. The depth of water in each container is the same.



In which container does the water exert the greatest pressure on the base of the container?

A. A

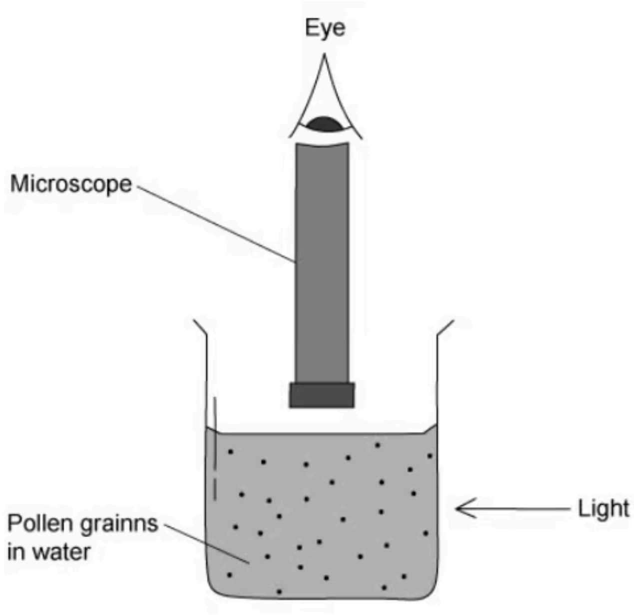
B. B

C. C

D. None, the pressure is the same in all three.

(1 mark)

14 In an experiment to determine the nature of liquids, some very tiny pollen grains are suspended in water. They are illuminated by a bright light, and observed under a microscope.



The view through the microscope is of small, bright dots, which move about at random, rapidly and they frequently change direction.

Which of the following would be the correct explanation for the observation through the microscope?

A. The bright dots are water molecules, being hit by other water molecules.

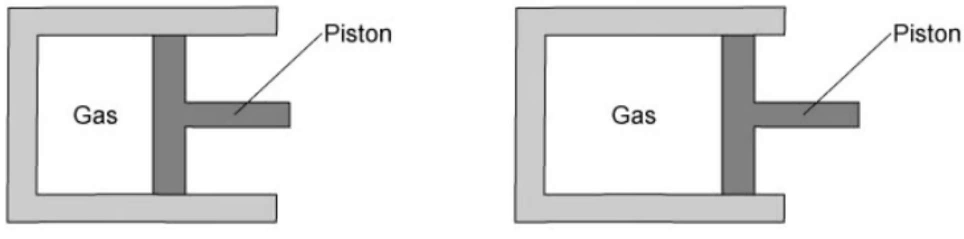
B. The bright dots are water molecules, which are vibrating due to their thermal kinetic energy.

C. The bright dots are pollen grains, which are being hit many times per second by water molecules.

D. The bright dots are pollen grains, which are vibrating due to their thermal kinetic energy.

(1 mark)

15 A sealed piston is used to expand a gas. No gas escapes during the process.



What happens to both the density of the gas and to its pressure when it is expanded in this way?

	Density	**Pressure**
A	stays the same	stays the same
B	increases	decreases
C	decreases	stays the same
D	decreases	decreases

(1 mark)

16 Which statement best describes a metal experiencing thermal expansion?

A. The size of the metal atoms increases

B. The space between the metal atoms increases

C. The metal atoms change state

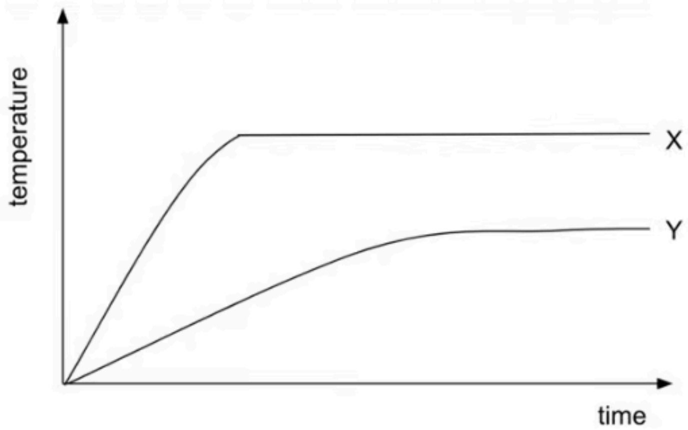
D. The metal atoms gain enough energy to overcome the intermolecular forces of attraction holding them in place

(1 mark)

17 Two beakers, X and Y, containing different liquids are heated by a student.

Both beakers contain the same mass of liquid, and are heated using the same heater set to the same power.

The graph shows how the temperature of each liquid changes with time.



What conclusion can be correctly drawn from the graph?

A. X has a higher melting point than Y.

B. Y boils sooner than X.

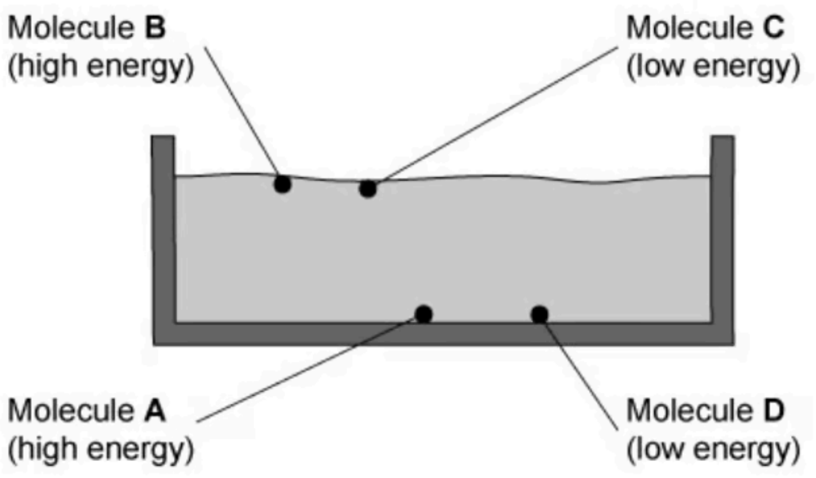
C. X has a lower specific heat capacity than Y.

D. Y has a higher boiling point than X.

(1 mark)

18 The diagram shows four water molecules in a swimming pool. It also shows how much energy each molecule has.

Which of the water molecules is most likely to *evaporate* from the liquid?



A. Molecule A

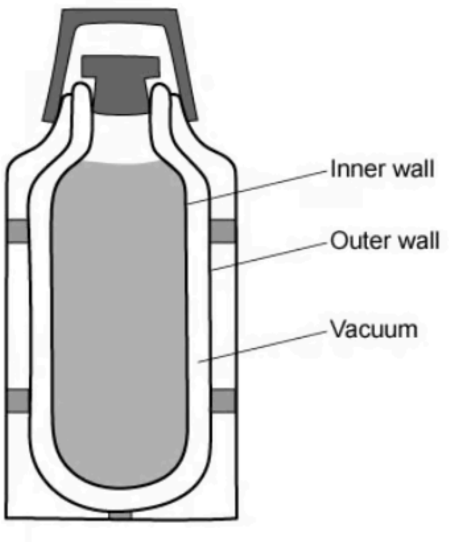
B. Molecule B

C. Molecule C

D. Molecule D

(1 mark)

19 The walls of a vacuum flask contain two layers of glass, separated by a vacuum.



Which method(s) of thermal energy transfer are prevented by the vacuum?

A. Radiation

B. Conduction and radiation

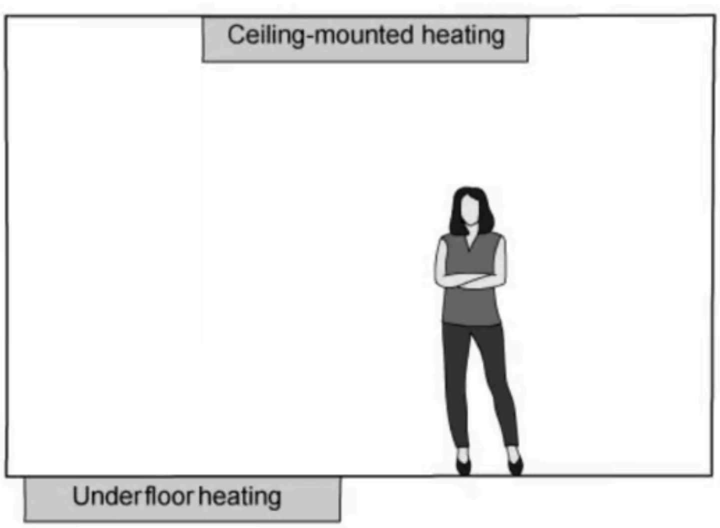
C. Convection and radiation

D. Conduction and convection

(1 mark)

20 In modern housing, it is convenient to locate heaters so they are not on the walls, taking up valuable wall space for furniture.

Other than wall-mounted radiators, two other options would be viable: ceiling heaters or underfloor heaters.



Which of the two options would heat the room more effectively and why?

	Option	Explanation
A	ceiling-mounted	hot air is more dense and falls
B	ceiling-mounted	hot air is less dense and falls
C	underfloor	hot air is less dense and rises
D	underfloor	hot air is more dense and rises

(1 mark)

21 What is the unit of wave speed?

A. Hertz

B. Metre

C. Second

D. Metres per second

(1 mark)

22 Water waves travel from deeper water to more shallow water. This causes them to refract.

Which wave property always remains the same when refraction occurs?

A. Wavelength

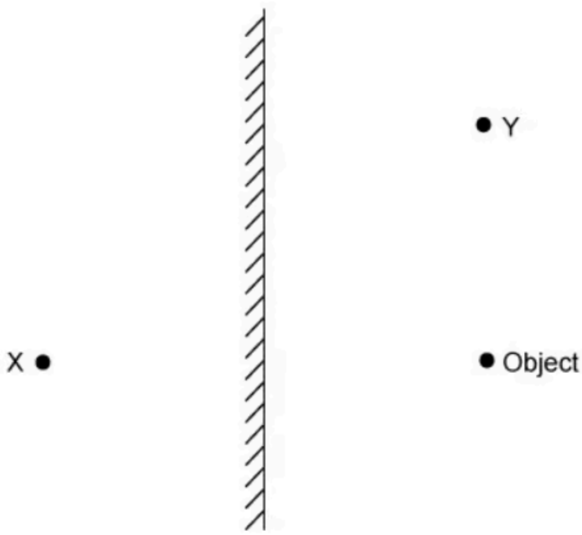
B. Frequency

C. Amplitude

D. Speed

(1 mark)

23



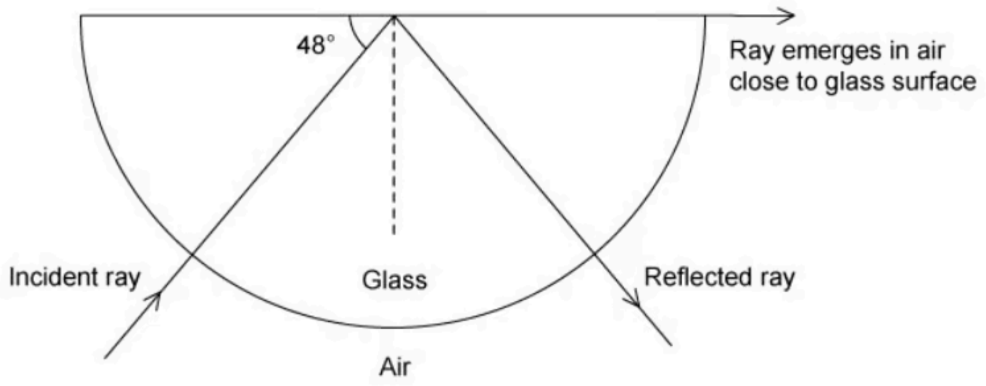
An object is placed in front of a mirror. Where is the image formed, and is the image real or virtual?

	Image location	Image type
A	X	real
B	X	virtual
C	Y	real
D	Y	virtual

(1 mark)

24 A ray of light is shone into a semi-circular glass block so that it makes an angle of

48° with the rear, flat edge of the block as shown.



Calculate the refractive index of the glass.

A. 0.023

B. 1.49

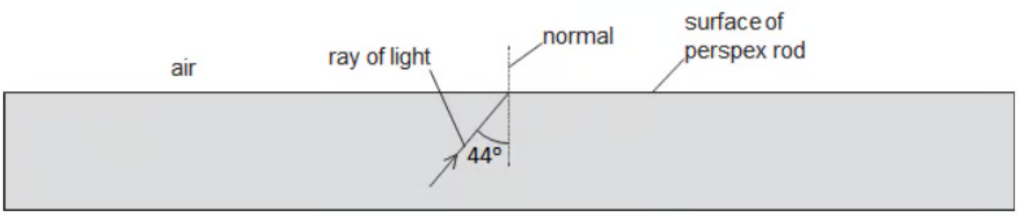
C. 2.38

D. 1.35

(1 mark)

25 The diagram shows a ray of light inside a perspex rod.

Perspex has a critical angle of 42°.



Which row in the table correctly states whether there is any light reflected from the surface, and whether there is any light refracted out of the perspex rod?

	Light reflected?	Light refracted?
A	yes	yes
B	yes	no
C	no	yes
D	no	no

(1 mark)

26 The table below shows the entire electromagnetic spectrum. It runs in order of increasing frequency.

Three parts of the spectrum are missing.

radio waves	**X**	infrared	visible light	**Y**	x-rays	**Z**

What are parts **X**, **Y** and **Z**?

	X	Y	Z
A	ultraviolet	microwaves	gamma rays
B	microwaves	gamma rays	microwaves
C	gamma rays	ultraviolet	microwaves
D	microwaves	ultraviolet	gamma rays

(1 mark)

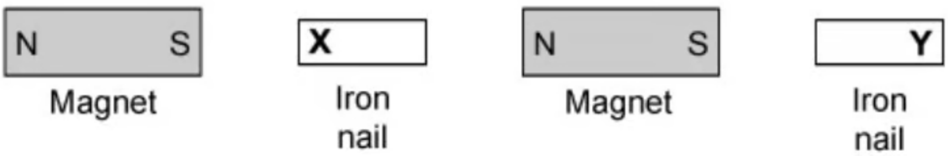
27 The table gives common uses of electromagnetic radiation.

Which row correctly matches the type of electromagnetic radiation with its use?

	Detecting banknote forgery	**TV remote controllers**	**Sterilising medical instruments**
A	microwaves	visible light	microwaves
B	infrared	infrared	γ-rays
C	ultraviolet	radio waves	infrared
D	ultraviolet	infrared	γ-rays

(1 mark)

28 A student lines up some permanent magnets with soft iron nails in between as shown in the diagram below.

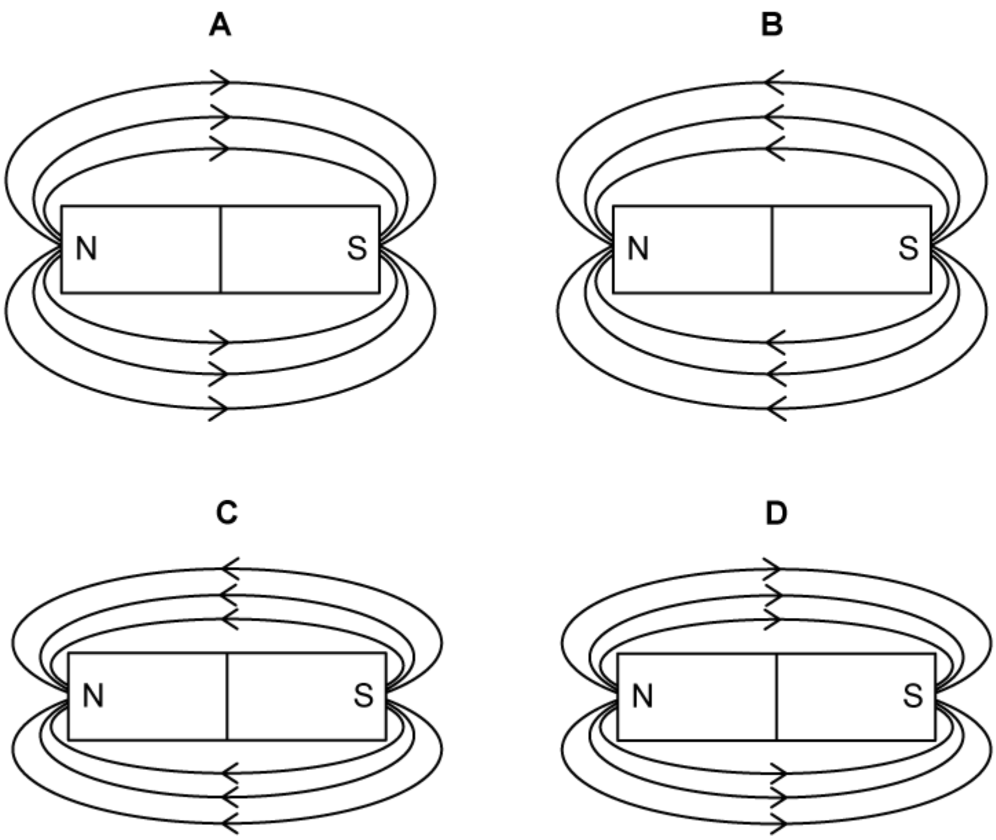


What would the magnetic poles of locations **X** and **Y** be?

	X	Y
A	north	north
B	north	south
C	south	north
D	south	south

(1 mark)

29 Which magnet has the strongest magnetic field?



(1 mark)

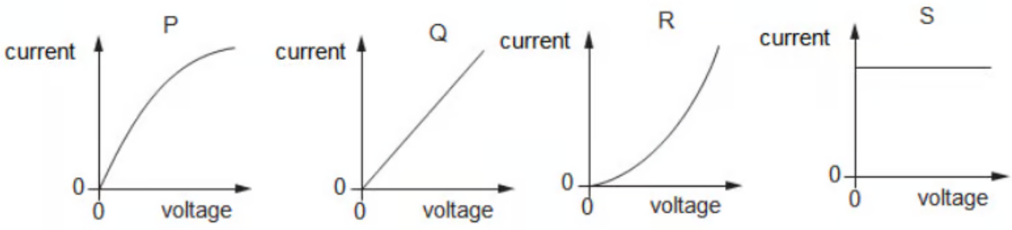
30 A student has four copper wires, of different dimensions.

Which of the following wires has the largest resistance?

	Length of wire / cm	Diameter of wire / mm
A	100	0.2
B	300	0.1
C	50	0.4
D	30	0.5

(1 mark)

31 Four current-voltage graphs are given below.



Which graph shows an ohmic resistor and which shows a filament lamp?

	Filament lamp	Ohmic resistor
A	Q	S
B	R	Q
C	P	Q
D	Q	R

(1 mark)

32 What is the definition of an electric field?

A. A region in space in which a mass experiences a force due to the Earth's mass.

B. A region in space through which electromagnetic radiation is passing.

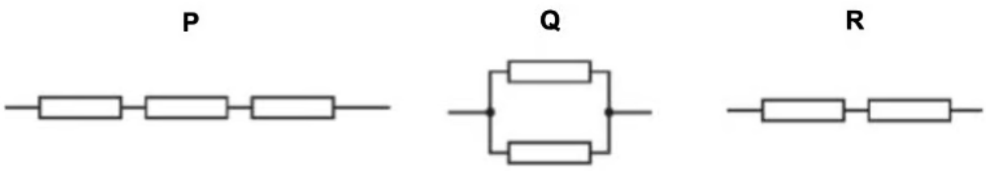
C. A region in space in which a compass needle experiences a force.

D. A region in space in which an electric charge experiences a force.

(1 mark)

33 A student connects together some identical resistors in three different combinations, as shown in the diagram.

Which of the answers below shows the correct order of the combinations from the lowest resistance to the highest?



A. P → Q → R

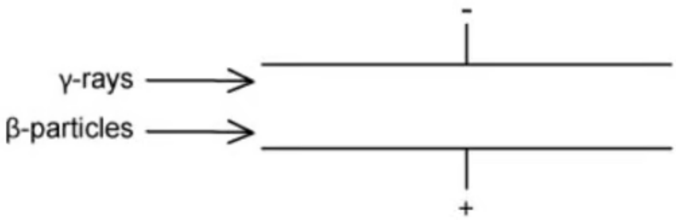
B. Q → P → R

C. Q → R → P

D. P → R → Q

(1 mark)

34 Beta and gamma radiation are passed through two metal plates, with equal and opposite charges, as shown in the diagram below.

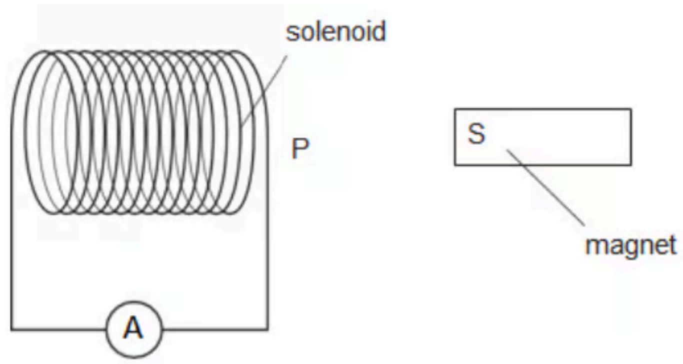


Which direction, if any, would the β-particles and γ-rays be deflected?

	β-particles	**γ-rays**
A	into the page	continue straight
B	towards the negative plate	out of the page
C	continue straight	towards the negative plate
D	towards the positive plate	continue straight

(1 mark)

35 The south pole of a magnet is brought near to a solenoid, which is connected to an ammeter. The magnet is then pulled away again.



As the magnet is moved towards point P the needle on the ammeter deflects. When it is moved away from point P, the needle on the ammeter also deflects.

What magnetic pole is produced at P when the south pole of the magnet is brought towards and away from the solenoid?

	S pole brought towards P	**S pole moved away from P**
A	N	N
B	N	S
C	S	N
D	S	S

(1 mark)

36 The number of turns in the primary coil and secondary coil of a transformer is N_p and N_s respectively.

Which of the following statements represents a step-up transformer?

A. $N_s = N_p$

B. $N_s > N_p$

C. $N_s < N_p$

D. $N_p > N_s$

(1 mark)

37 Which of the following statements would **not** be correct for the nucleus of **any** atom?

A. The nucleus contains protons and neutrons.

B. The nucleus has a total charge of zero.

C. The nucleus is very small compared to the size of the atom.

D. The atom contains the same number of protons and electrons.

(1 mark)

38 A radioactive nucleus emits a **β**-particle.

What happens to the proton number and the nucleon number of this nucleus?

	Proton number	**Nucleon number**
A	increases by 1	stays the same
B	stays the same	decreases by one
C	decreases by 2	decreases by 4
D	decreases by 1	stays the same

(1 mark)

39 Which two elements are the most commonly found in the Sun?

A. Hydrogen and oxygen

B. Helium and carbon

C. Hydrogen and helium

D. Nitrogen and oxygen

(1 mark)

40 The orbit of the Earth around the Sun can be thought of as following a circular path with a circumference of 942×10^6 km.

What is the approximate orbital speed of the Earth?

A. 10 000 km /h

B. 100 000 km /h

C. 2.5×10^6 km /h

D. 50×10^6 km /h

(1 mark)